Wireshark Level 1 – Network Traffic Analysis

This project demonstrates the installation and use of Wireshark for basic network traffic analysis. Wireshark was installed on Ubuntu using the apt package manager. After installation, packet capture was initiated on the eth0 interface. Network traffic was generated using the ping command to send ICMP requests to an external server. Captured packets were analyzed in real time using Wireshark's interface. A display filter (icmp) was applied to isolate ICMP traffic, allowing clear observation of Echo Requests and Echo Replies. Detailed inspection of packets revealed multiple protocol layers including Frame, Ethernet, IP, and ICMP. Key fields such as source/destination IP, TTL, packet type, checksum, and sequence numbers were examined. This project builds foundational skills in network monitoring, packet filtering, and protocol analysis, which are essential for cybersecurity roles such as network defense and incident response.

Real-World Context

ICMP traffic is widely used in real-world networking for diagnostics such as connectivity testing. However, it is also commonly leveraged in cybersecurity contexts. Attackers may use ICMP for reconnaissance to identify active hosts on a network, or for more advanced techniques such as ICMP tunneling, where data is covertly transmitted within ICMP packets to bypass firewall restrictions.

Figure 1: Live Packet Capture

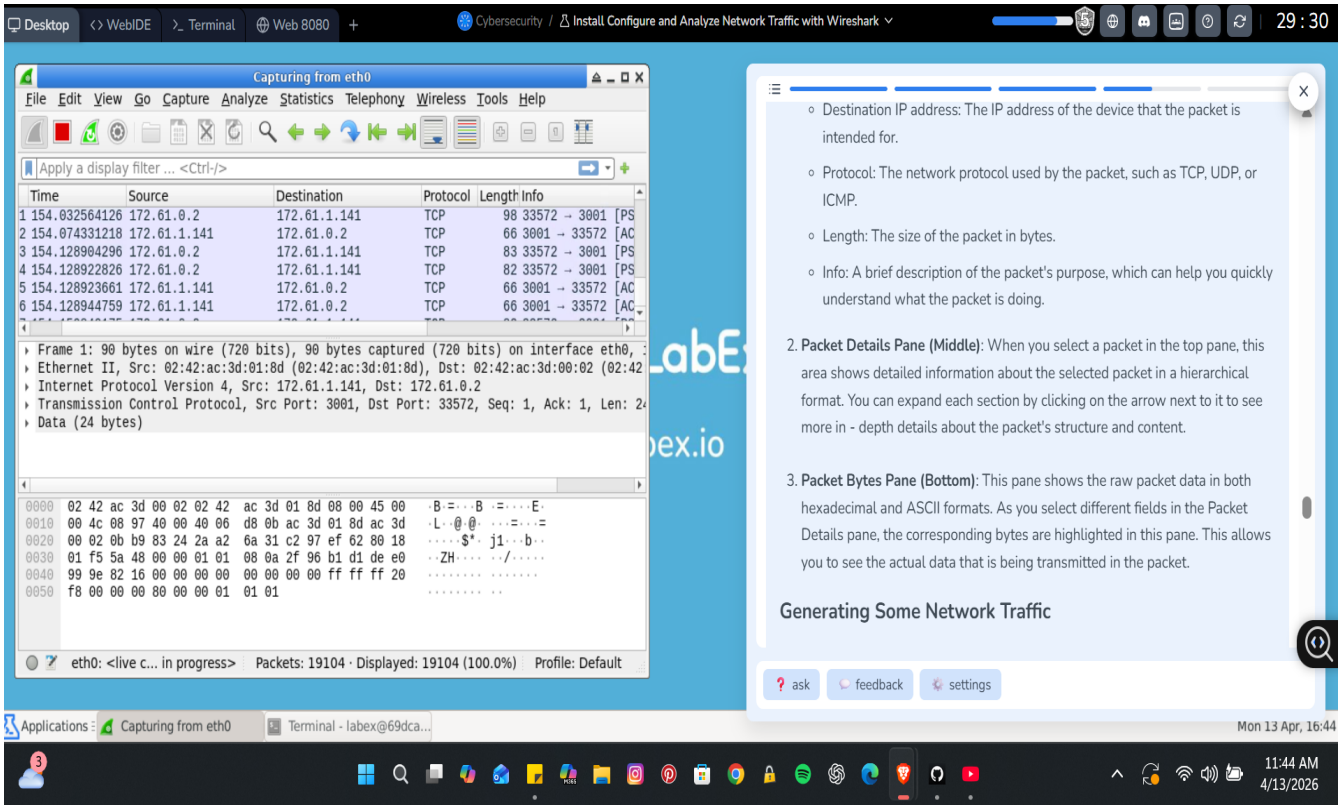
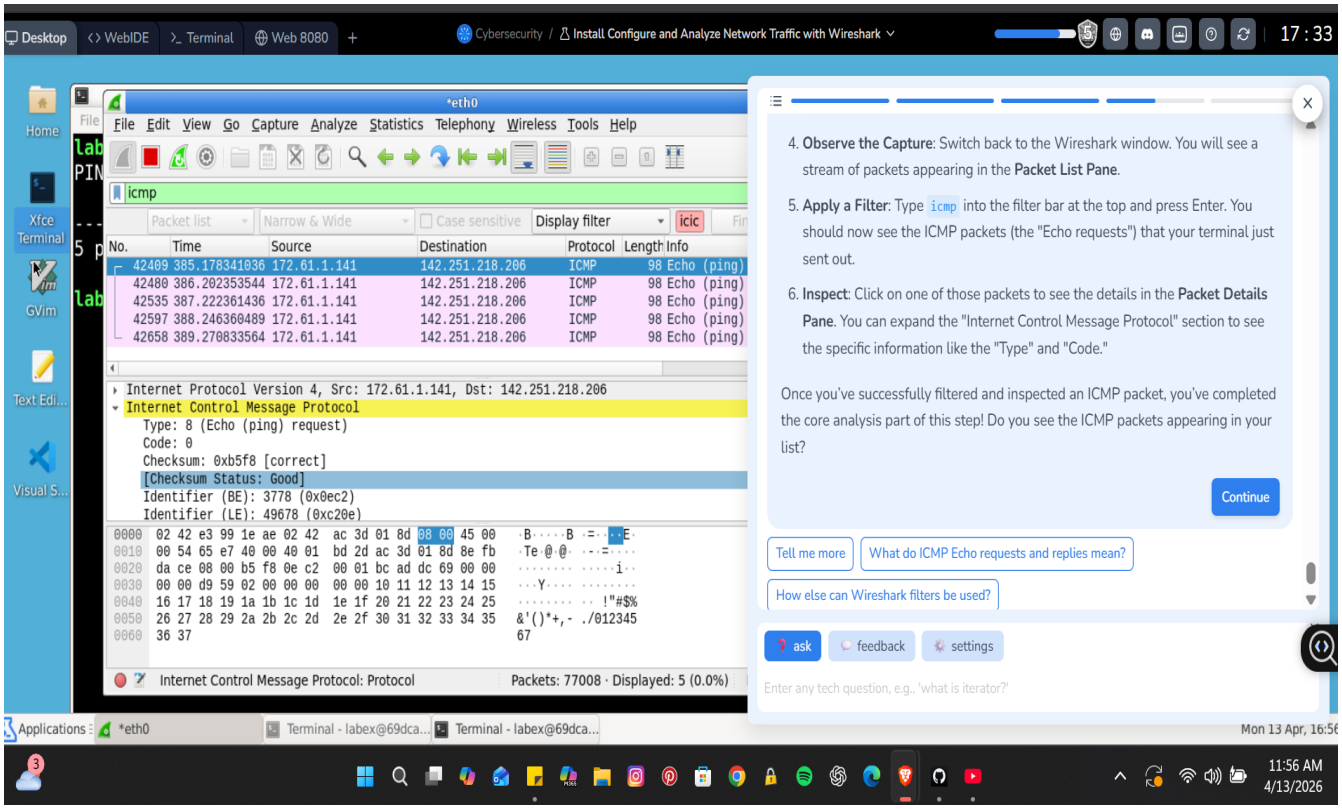


Figure 2: ICMP Packet Filtering and Analysis



Potential Security Implications

Although ICMP is essential for network diagnostics, abnormal or excessive ICMP traffic can indicate potential security threats. For example, a high volume of ICMP requests may suggest a ping flood attack, which is a form of Denial-of-Service (DoS). Additionally, unusual ICMP patterns could signal unauthorized scanning activity or covert communication channels used by malware.

What I Learned

This project enhanced my understanding of how network traffic flows across different protocol layers. I learned how to capture live packets, apply filters to isolate specific protocols, and analyze packet details such as IP addressing and ICMP structure. It also provided practical insight into how these skills are applied in real-world cybersecurity scenarios such as traffic monitoring and threat detection.